

AI Applications Portfolio

Welcome to my AI Applications Portfolio. I'm Andrew Kuruvilla, an aspiring AI professional with extensive hands-on experience across multiple domains including Azure AI services, natural language processing, computer vision, and industry-specific applications.

This presentation showcases my expertise through course projects and assignments, demonstrating my ability to contribute to innovative, ethical, and impactful AI-driven solutions across diverse industries from healthcare to space exploration.

A by Andrew Kuruvilla

Azure AI Document Intelligence

Project Overview

Explored Azure AI Document Intelligence to extract structured data from receipts using prebuilt models in the Azure AI Foundry portal.

Skills Acquired

Created and managed AI resources in Azure; distinguished OCR from structured data extraction; analyzed confidence scores for extracted fields.

Tools Mastered

Azure AI Foundry Portal, Azure AI Document Intelligence, cloud resource management.

Outcome

Successfully extracted key receipt details, demonstrating automation potential for retail and finance sectors.

NASA Space Mission AI Project



Project Description

Designed an AI-based autonomous navigation system for a Mars rover, focusing on real-time path planning and obstacle avoidance using machine learning and sensor fusion.



Learning Outcomes

Developed AI architectures for computer vision, reinforcement learning, and sensor fusion; tested AI systems in simulated environments.



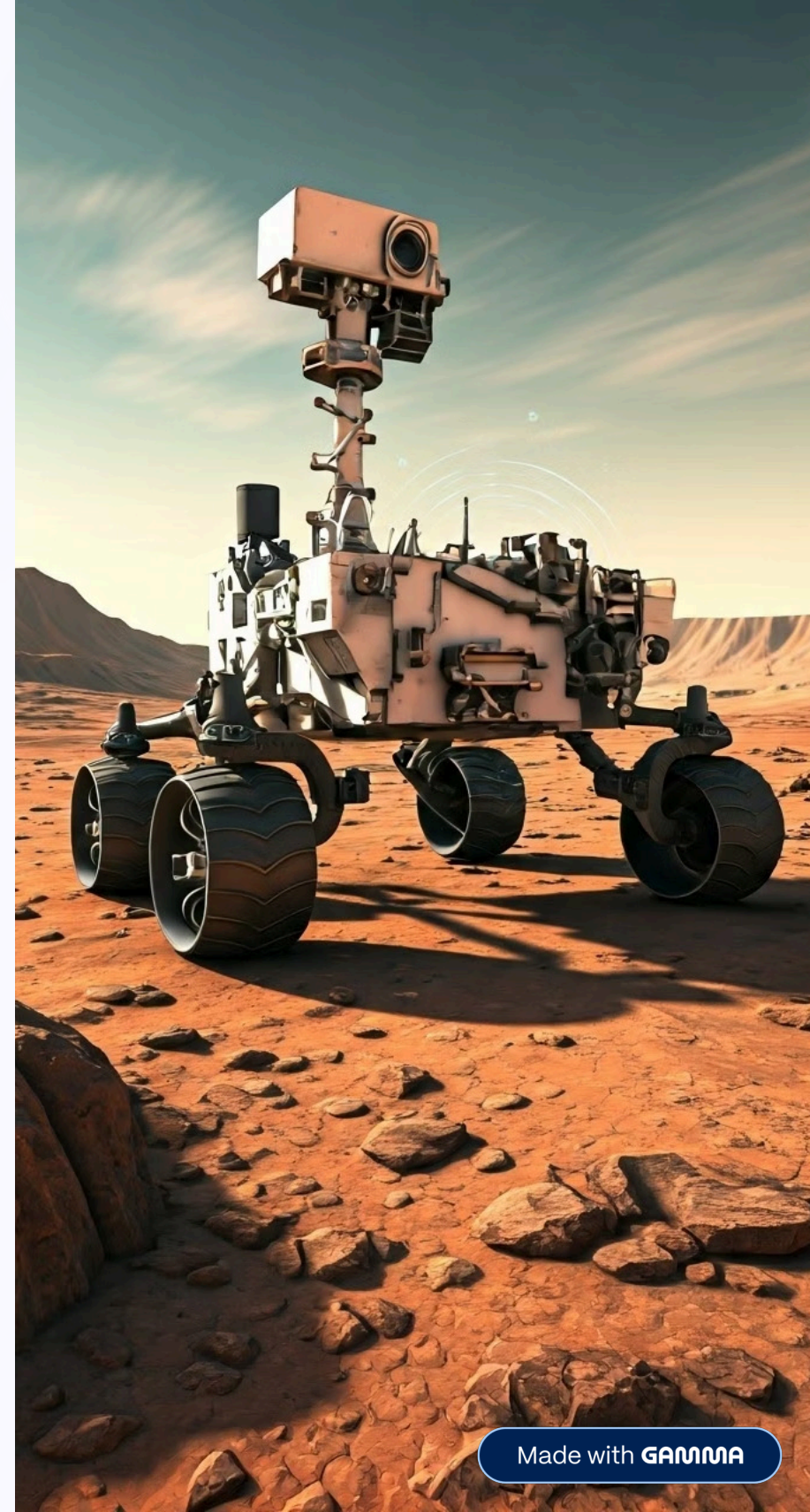
Technologies Used

Convolutional Neural Networks (CNNs), Deep Q-Networks (DQNs), ROS, Gazebo, LiDAR, OpenCV.



Project Results

Proposed a robust system to enhance Mars rover navigation, validated through simulated testing scenarios.



AI in Fraud Detection & Smart Cities

Fraud Detection Case Study

Analyzed AI-driven fraud detection across seven financial institutions, focusing on technologies, benefits, and challenges.

Learned about machine learning, deep learning, and behavioral analytics for fraud detection while addressing data privacy and transparency concerns.

Produced a comprehensive report on AI's role in reducing fraud losses, applicable globally.

Smart Cities Analysis

Compiled notes on AI applications in smart cities, including crime prediction, waste management, and public transport optimization.

Explored predictive policing, smart waste sorting, and transit scheduling; identified challenges like implementation cost and data accuracy.

Gained valuable insights into AI-driven urban management for enhanced sustainability and efficiency.



AI in Government & Public Sector

Disaster Response Analysis

Examined AI's role in disaster response for the Moroccan earthquake, utilizing satellite imagery analysis and deep learning models.



Privacy & Bias Considerations

Addressed critical concerns regarding data privacy and algorithmic bias in government AI applications.



Citizen Engagement Platform

Proposed an AI-driven citizen engagement platform for smart cities to enhance communication between residents and government.



Outcome & Impact

Demonstrated AI's potential to enhance governance and proposed a scalable solution for urban transparency.



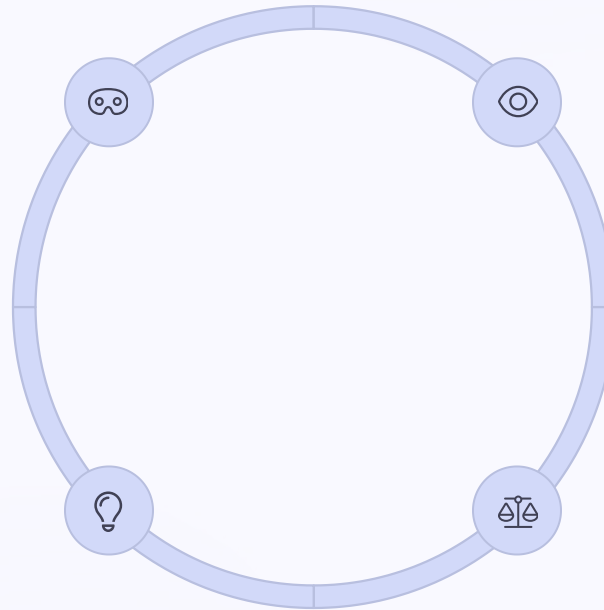
Ethical Implications of AI

Deepfake Misuse

Analyzed deepfake technology using generative adversarial networks and its potential for misuse in entertainment.

Explainable AI

Proposed ethical frameworks and explainable AI solutions for responsible deployment.



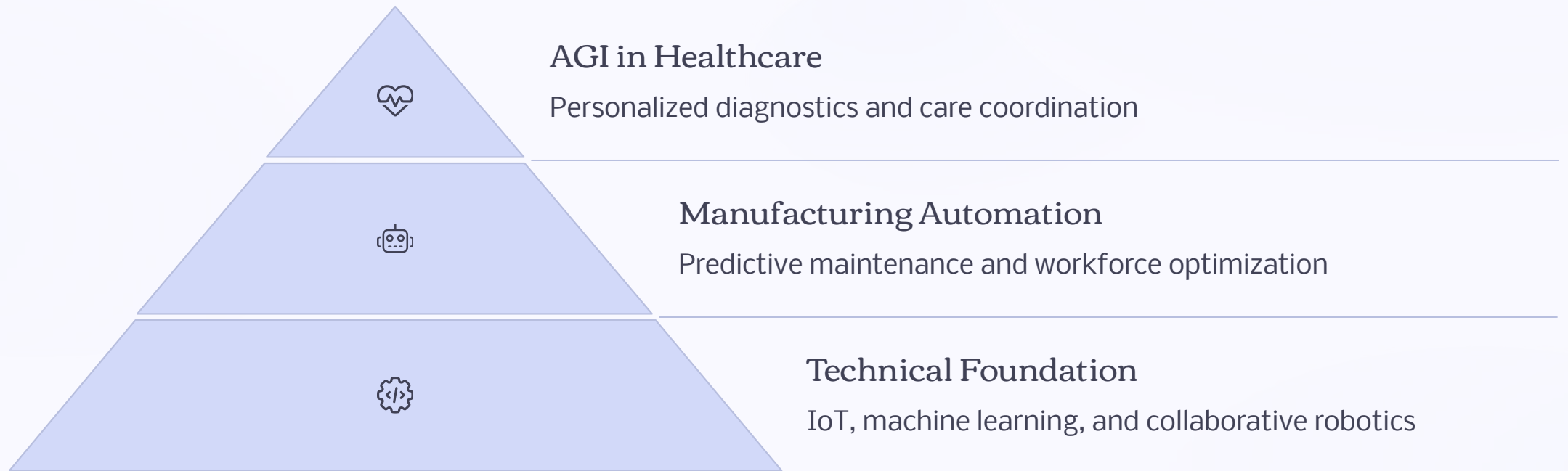
Privacy Concerns

Explored AI surveillance implications and compliance with regulations like GDPR.

Algorithmic Bias

Investigated biased algorithms and their impact on marginalized communities.

AI in Manufacturing & Healthcare



In manufacturing, I studied AI-driven predictive maintenance at General Electric and proposed an AI workforce management system using collaborative robotics and computer vision, highlighting AI's role in reducing downtime and enhancing efficiency.

For healthcare, I proposed AGI applications for personalized diagnostics, care coordination, and global health equity, demonstrating AGI's cognitive flexibility for holistic healthcare solutions while addressing privacy concerns.



Applications & Contact Information

6+

Industries

Finance, urban planning,
entertainment, manufacturing,
healthcare, aerospace

10+

Projects

Spanning from document
intelligence to space exploration

15+

AI Tools

Including Azure AI, Hugging Face,
ROS, and OpenCV

I'm eager to apply my diverse AI expertise to solve complex challenges, promote ethical technology use, and drive innovation across industries. My skills include Azure AI Vision, Language, and Content Safety Studio, plus expertise in ethical AI, blockchain, and regulatory compliance.

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